
Floating and Positioning

Topics

Floating elements to the left and right

Clearing floated elements

Containing floated elements

Creating text-wrap shapes

Relative positioning

Absolute positioning and containing blocks

Fixed positioning

Floating and Positioning

Floating an element **moves** it to the **left or right** and allows the following text to wrap around it

Positioning is a way to **specify the location of an element** anywhere on the page with pixel precision

Normal Flow

In the CSS layout model, **text elements are laid out from top to bottom in the order in which they appear in the source**, and **from left to right** in left-to-right reading languages

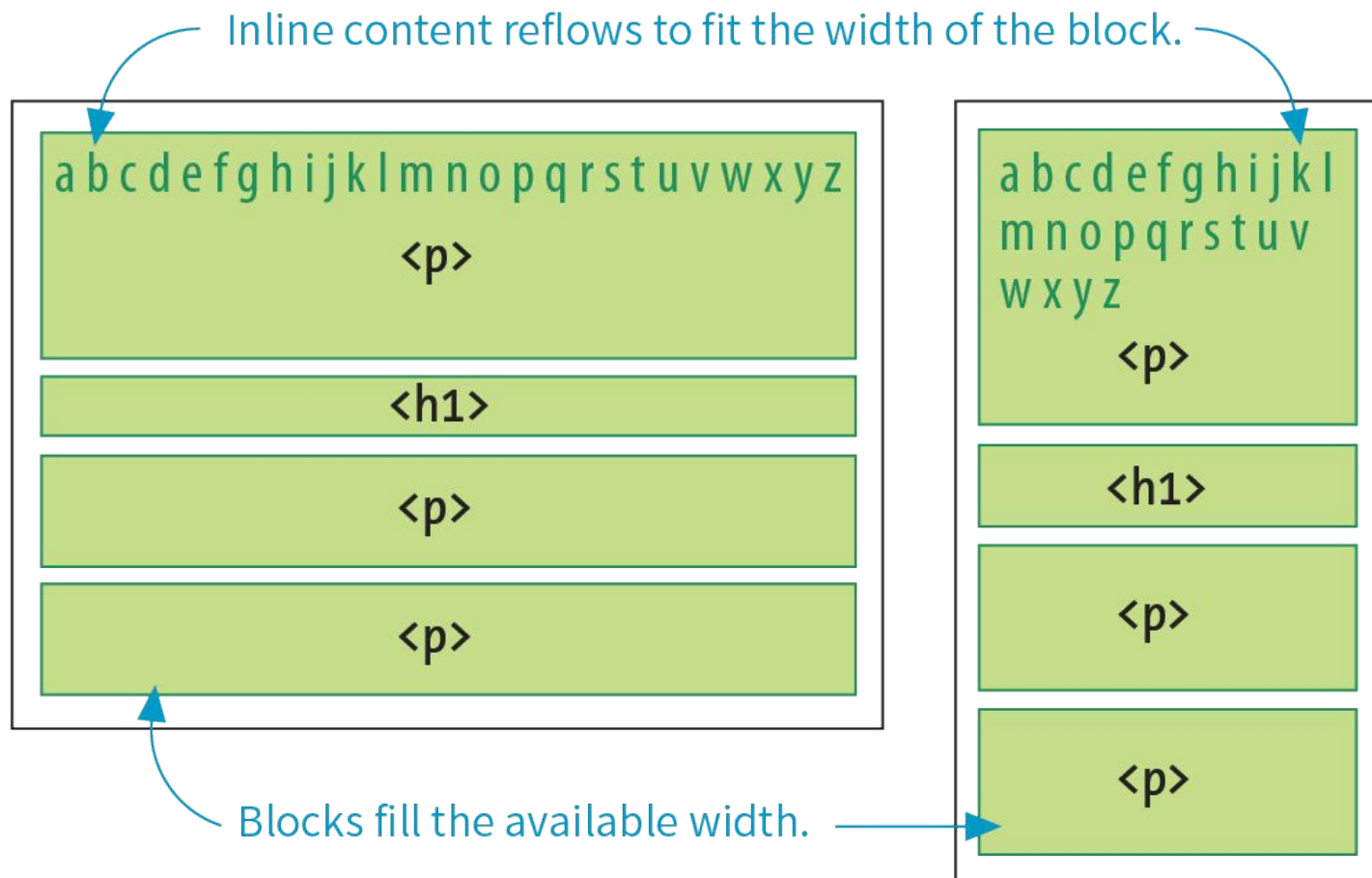
Block elements stack up on top of one another and **fill the available width** of the browser window or other containing element

Inline elements and text characters **line up next to one another to fill the block elements**

Objects in the normal flow affect the layout of the objects around them

Blocks are laid out in the order in which they appear in the source.

Each block starts on a new line.



Floating

The **float** property moves an element as far as possible to the left or right, allowing the following content to wrap around it

float

Values: `left` | `right` | `none`

Default: `none`

Floating Example

```
img {  
  float: right;  
}
```

```
<p> After the cream is  
frozen rather stiff,... </p>
```

Floating Example

Inline image in the normal flow



Space next to image is held clear

After the cream is frozen rather stiff, prepare a tub or bucket of coarsely chopped ice, with one-half less salt than you use for freezing. To each ten pounds of ice allow one quart of rock salt. Sprinkle a little rock salt in the bottom of your bucket or tub, then put over a layer of cracked ice, another layer of salt and cracked ice, and on this stand your mold, which is not filled, but is covered with a lid, and pack it all around, leaving the top, of course, to pack later on. Take your freezer near this tub. Remove the lid from the mold, and pack in the cream, smoothing it down until you have filled it to overflowing. Smooth the top with a spatula or limber knife, put over a sheet of waxed paper and adjust the lid.

Inline image floated to the right

Image moves over, and text wraps around it

After the cream is frozen rather stiff, prepare a tub or bucket of coarsely chopped ice, with one-half less salt than you use for freezing. To each ten pounds of ice allow one quart of rock salt. Sprinkle a little rock salt in the bottom of your bucket or tub, then put over a layer of cracked ice, another layer of salt and cracked ice, and on this stand your mold, which is not filled, but is covered with a lid, and pack it all around, leaving the top, of course, to pack later on. Take your freezer near this tub. Remove the lid from the mold, and pack in the cream, smoothing it down until you have filled it to overflowing. Smooth the top with a spatula or limber knife, put over a sheet of waxed paper and adjust the lid.



Floating Inline and Block elements

It is possible to float any HTML element, both inline and block-level

Floating an inline text element

It is necessary to specify the **width** for floated text elements

Floated inline elements behave as block elements

Once you float an inline element, it follows the display rules for block-level elements, and margins are rendered on all four sides

Confirm this property using Chrome DevTool

Margins on floated elements do not collapse

Exercise

Create an `index.html` file and add a basic html structure

Add external stylesheet named `style.css`

Add an `img` tag with an image inside the body tag

Add a `p` tag with lorem ipsum content

Observe where the image is located

Float the image by adding a `float: left` or `float:right` rule to the `img` tag

Exercise

Surround a word inside the `p` using the `span` tag and give the `span` tag width, height, and margin

Check if the width, height, margin rules have effect on the `span` element

Float the `span` by adding a `float: left` or `float: right` rule to the `span` tag

Check if the width, height, margin rules have effect on the `span` element

Floating block elements

You must provide a **width** for floated block elements

Elements do not float higher than their reference in the source

Clearing Floated Elements

Turn the text wrapping off

Applying the **clear** property to an element prevents it from appearing next to a floated element and forces it to start against the next available “clear” space below the float

clear

Values: `left` | `right` | `both` | `none`

Default: `none`

Exercise

Clear the text on the left or on the right of a floated element

Positioning

Elements can be positioned

- relative to where they would normally appear in the flow

- at a particular spot on the page

- relative to the viewport

Types of Positioning

position

Values: `static` | `relative` | `absolute` | `fixed`

Default: `static`

Specifying Position

top, right, bottom, left

Values: `length` | `percentage` | `auto`

Default: `auto`

The value provided for each offset property defines the distance the element should be moved away from that respective edge

Relative Positioning

Moves an element relative to its original spot in the flow

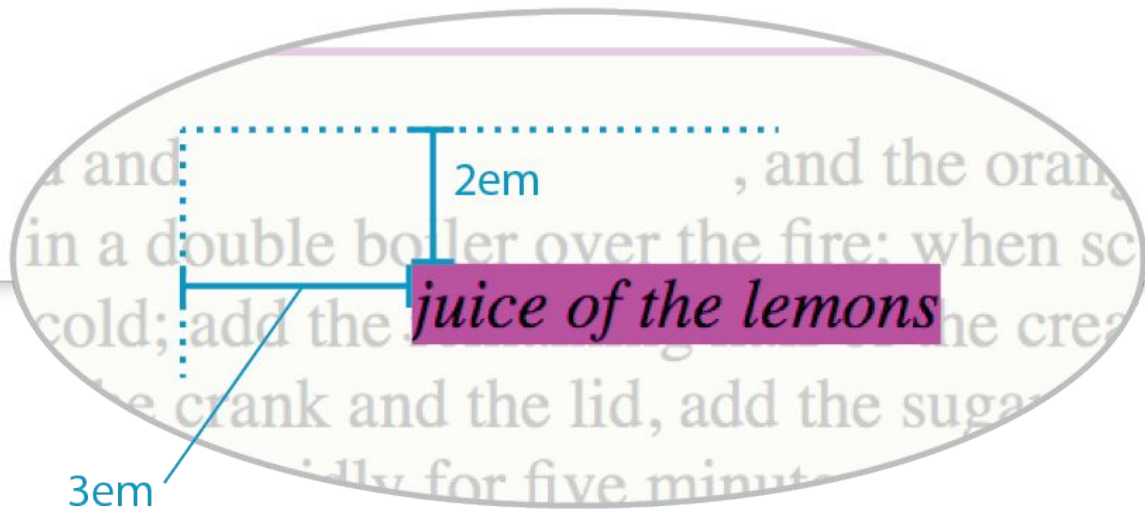
The space it would have occupied is preserved and continues to influence the layout of surrounding content

Overlap happens

Relative Positioning Example

```
em {  
  position: relative;  
  top: 2em; /* moves element down */  
  left: 3em; /* moves element right */  
  background-color: fuchsia;  
}
```

Mix the sugar, the grated rind and _____, and the orange juice together. Put half the cream in a double boiler over the fire: when scalding hot, stand it aside until perfectly cold; add the *juice of the lemons* the cream and freeze it rather hard. Remove the crank and the lid, add the sugar mixture, replace the lid and crank, and turn rapidly for five minutes; repack to ripen.



Exercise

Surround a word in the paragraph using `em` tag

Add the following rules

```
em {  
  position: relative;  
  top: 2em; /* moves element down */  
  left: 3em; /* moves element right */  
  background-color: fuchsia;  
}
```

Absolute Positioning

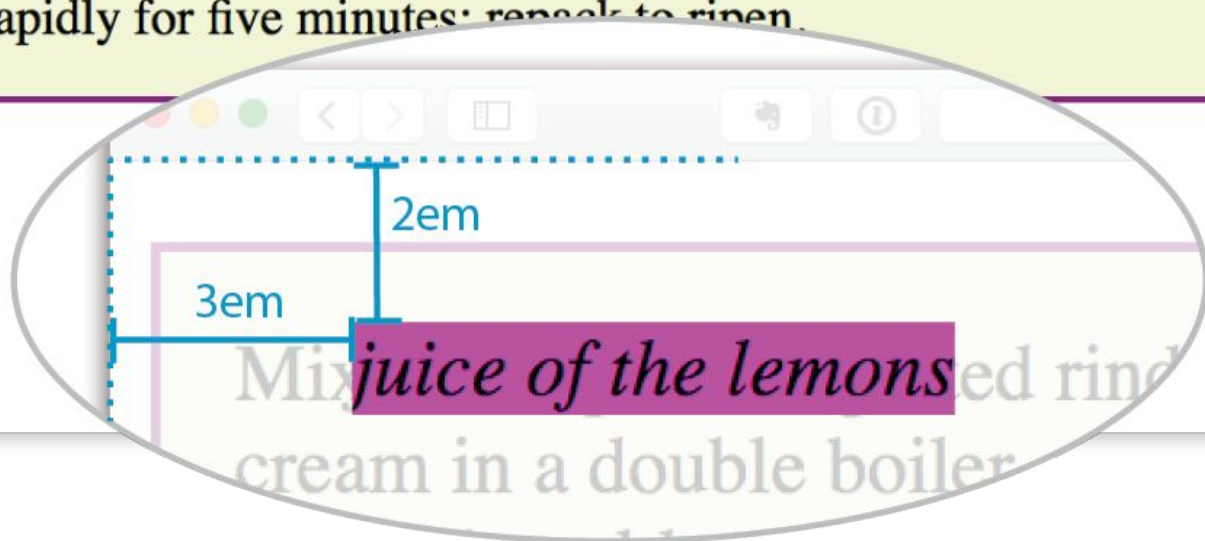
Absolutely positioned elements have no influence on the layout of surrounding elements

In absolute positioning the element is positioned relative to its nearest containing block

Absolute Positioning Example

```
em {  
  position: absolute;  
  top: 2em; /* moves element down */  
  left: 3em; /* moves element right */  
  background-color: fuchsia;  
}
```

Mix *juice of the lemons*ed rind and , and the orange juice together. Put half the cream in a double boiler over the fire; when scalding hot, stand it aside until perfectly cold; add the remaining half of the cream and freeze it rather hard. Remove the crank and the lid, add the sugar mixture, replace the lid and crank, and turn rapidly for five minutes: ~~repeat to ripen.~~



Exercise

Use the following css rules for the `em` element

```
em {  
  position: absolute;  
  top: 2em; /* moves element down */  
  left: 3em; /* moves element right */  
  background-color: fuchsia;  
}
```

Containing Blocks

The position and size of an element's box(es) are sometimes computed relative to a certain rectangle, called the containing block of the element

The most common way to make an element into a containing block is to set its position to relative, but not move it with any offset values

Containing Blocks

Determining the containing block of an element

If the positioned element is not contained within another positioned element, then it will be placed relative to the initial containing block (created by the html element)

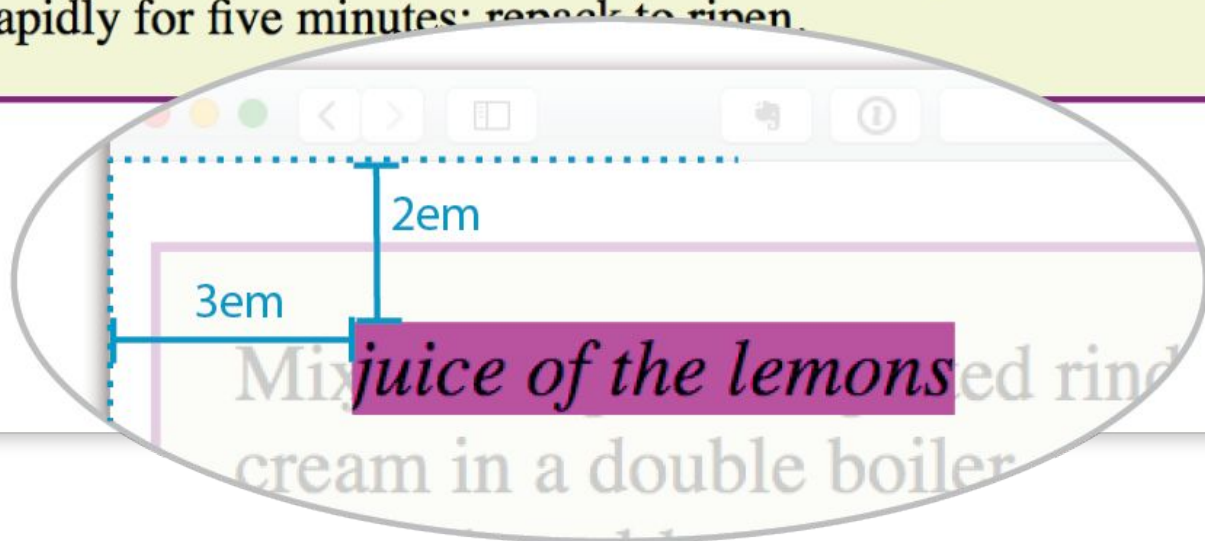
But if the element has an ancestor (i.e., is contained within an element) that has its position set to relative, absolute, or fixed, the element will be positioned relative to the edges of that element instead

Containing Blocks

The next two slides show the contrast between the two

In the first case the paragraph where the em element resides is not positioned element but the second paragraph is positioned element

Mix *juice of the lemons*ed rind and , and the orange juice together. Put half the cream in a double boiler over the fire; when scalding hot, stand it aside until perfectly cold; add the remaining half of the cream and freeze it rather hard. Remove the crank and the lid, add the sugar mixture, replace the lid and crank, and turn rapidly for five minutes: ~~repeat to ripen.~~



Mix the sugar, the grated rind and , and the orange juice together. Put half the cream *juice of the lemons* over the fire; when scalding hot, stand it aside until perfectly cold; add the remaining half of the cream and freeze it rather hard. Remove the crank and the lid, add the sugar mixture, replace the lid and crank, and turn rapidly for five minutes; repack to ripen.

Mix the sugar, the grated rind
cream *juice of the lemons* over
perfectly cold; add the

Exercise

Add one more paragraph above the previous paragraph

Add **position: relative** to paragraph

Check what happens to the **em** element position

Absolute Positioning

The offset values apply to the outer edges of the element box (the outer margin edge) for absolutely positioned elements

For relatively positioned elements, the offset is measured to the box itself (not the outer margin edge)

Absolutely positioned elements always behave as block-level elements

For example, the margins on all sides are maintained, even though this is an inline element

It also permits a width to be set for the element

Stacking Order

By default, elements stack up in the order in which they appear in the document, but you can change the stacking order with the **z-index** property

Picture the **z-axis** as a line that runs perpendicular to the screen/page

z-axis

Values: `number` | `auto`

Default: `auto`

Stacking Order

The higher the number, the higher the element will appear in the stack

Lower numbers and negative values move the element lower in the stack

An example is shown in the next slide



By default, elements later in the document source order stack on top of preceding elements.

z-index: 100



z-index: 5

z-index: 1

You can change the stacking order with the z-index property. Higher values stack on top of lower values.

Fixed Positioning

Fixed positioning works just like absolute positioning

The significant difference is that the offset values for fixed elements are always relative to the viewport

The fixed positioned element stays put even when the rest of the page scrolls

Fixed elements are often used for menus that stay in the same place at the top, bottom, or side of a screen so they are always available, even when the content scrolls

Exercise

Add 6 paragraphs with lorem content

Add a div which has the following css rules

- 100px width and height

- background-color of lightblue

- fixed position to the right 0px and top 50vh

Check the effect by changing the viewport height and scrolling up and down